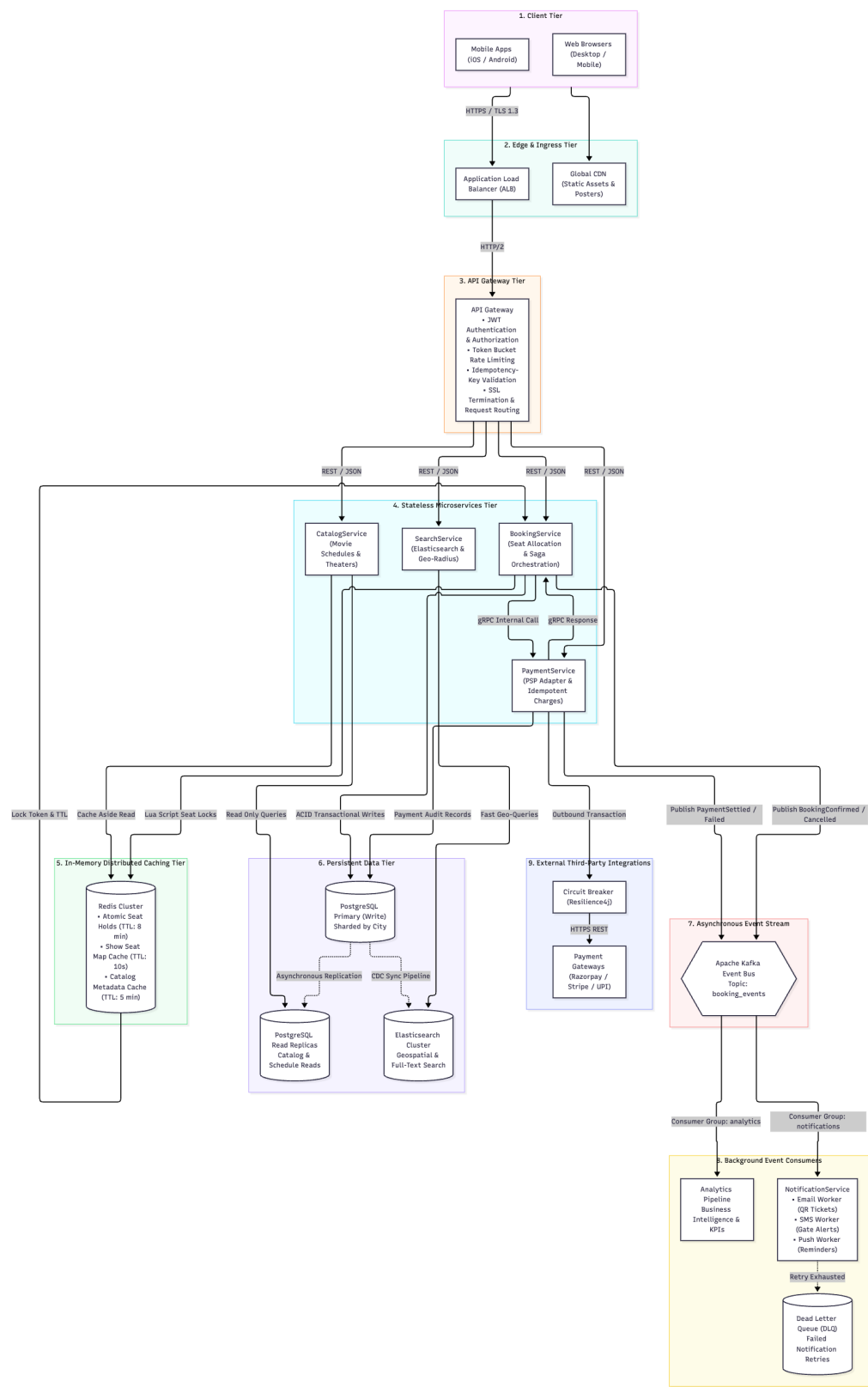


High-Level Architecture: Distributed System Topology

Core Architecture: Layered distributed topology spanning Edge/Ingress, Stateless Microservices, Redis Distributed Locking, City-Sharded PostgreSQL, Elasticsearch Geo-Discovery, and Kafka Event Bus.

Key Guarantees: Zero double bookings via atomic Redis locks (CP), 99.95% catalog uptime (AP), and decoupled async notifications.



System Component Roles & Communication Protocols

Detailed mapping of system tiers, microservice responsibilities, storage mechanisms, and inter-service communication protocols.

Component / Layer	Primary Role & Responsibilities	Technology & Pattern	Inter-Service Protocol
Edge & CDN	Caches posters, trailers, venue layouts; protects origin.	Cloudflare / AWS CloudFront	HTTPS / Anycast DNS
API Gateway	JWT auth, rate limiting (Token Bucket), idempotency verification.	Kong / Envoy Gateway	HTTP/2 TLS 1.3 Ingress
BookingService	Orchestrates seat holds, checkout lifecycle, and final booking confirmation.	Go / Node.js Microservice	REST (Client), gRPC (Payment), Kafka (Pub)
PaymentService	Idempotent payment processing, PSP gateway integration, and webhooks.	Java / Spring Boot	gRPC (Internal), HTTPS (External PSP)
SearchService	Geospatial radius queries, faceted filtering (language, format, genre).	Elasticsearch / OpenSearch	REST / JSON Queries
CatalogService	Serves movie metadata, theater details, and screening schedules.	Stateless App Pods	REST / Cache-Aside Read
In-Memory Tier	Atomic seat locks (TTL: 8 min) and short-lived seat layout caches.	Redis Cluster (Master-Replica)	RESP Protocol / Lua Scripts
Database Shards	ACID transaction persistence sharded horizontally by city_id.	PostgreSQL Primary + Read Replicas	SQL Connection Pool (PgBouncer)
Event Bus	Decouples post-booking operations (email, SMS, analytics, loyalty).	Apache Kafka (booking_events)	Binary TCP Protocol
NotificationService	Dispatches QR-coded tickets, SMS alerts, and mobile push notifications.	Consumer Worker Pool + DLQ	Kafka Consumer Group
External PSPs	Third-party payment settlement (Razorpay, Stripe, UPI).	Resilience4j Circuit Breaker	HTTPS REST + Async Webhooks

Architectural Reliability Highlights:

- **Concurrency Defense:** Redis distributed locks guarantee single-owner seat holds with automatic TTL expiration, preventing database locking overhead.
- **Zero Cross-Shard Joins:** Sharding PostgreSQL by city_id guarantees all cinema, screen, and show data for a transaction resides on one local shard.
- **Fault Isolation:** External payment gateway calls are protected with circuit breakers; background notification failures route to DLQs without degrading checkout latency.